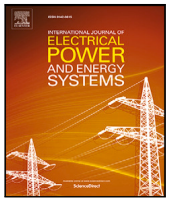
Electrical Power and Energy Systems 153 (2023) 109323



Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes





Model-free inertia estimation in bulk power grids through O-splines

Diego Rodales [a], Alejandro Zamora-Mendez [a,*], José Antonio de la O Serna [b], Juan M. Ramirez [c], Mario R. Arrieta Paternina [d], Lucas Lugnani [e], Gabriel E. Mejia-Ruiz [d], Alexander Sanchez-Ocampo [c], Daniel Dotta [e]

[a] University of Saint Nicholas of Hidalgo, Morelia, 58030, MICH, Mexico
[b] Autonomous University of Nuevo Leon, Monterrey, 66450, NL, Mexico
[c] Centre for Research and Advanced Studies, Guadalajara, 45019, JAL, Mexico
[d] National Autonomous University of Mexico, Mexico City, 04510, MEX, Mexico
[e] University of Campinas, Campinas, 13083-852, Brazil

ARTICLE INFO

Keywords:
Inertia estimation
Rate of change of frequency (RoCoF)
O-splines
Discrete-time Taylor–Fourier transform

ABSTRACT

This paper proposes a novel methodology to precisely estimate the generation units' inertia constant by using the wide-area monitoring systems (WAMS) information of bulk power grids. The method accurately estimates the rate of change of frequency (RoCoF) and the amplitude of active power deviations through the multivariate O-splines-based differentiators of the discrete-time Taylor–Fourier transform (DTTFT). The proposal utilises active power and frequency measurements that are synchronously collected from WAMS after load changes or generation trips. In the first stage, the Teager–Kaiser energy operator (TKEO) detects the right disturbance instant. Then, the first-order and zero-order O-splines differentiators are applied to the frequency and active power deviation measurements to simultaneously estimate the RoCoF and the amplitude of active power deviations in a sole stage. Thus, the inertia constants of all generation units derived from the second Newton law are estimated using all time-synchronised signals and processing in an exclusive step by the O-splines of the DTTFT in a short time. The attained results and their comparison with a state-of-the-art technique confirm the effectiveness and performance of the proposed strategy for inertia estimates, where simulated signals of two well-known benchmark power grids are tested.

1. Introduction

The active changes in modern power grids have imposed significant challenges on the provision of frequency control, since they have impacted the inertial response traditionally provided by the rotor masses of generation units along the power system. As consequence, power grids have also been exposed to imbalances that alter the reaction of all primary, secondary and supplementary control actions. These problems together with the paradigm about the uncertain knowledge of the total grid inertia [1], advocate the need of disposing of a methodology that estimates the grid inertia, mainly without requiring model information that may be outdated or poorly tuned.

Fortunately, modern grids are equipped with wide-area monitoring systems (WAMS) that provide operators with synchronised large data streams and can be used for grid inertial estimation. Methods that use the WAMS information are model-free (or data-driven), and they require little operator intervention for acquiring assessments, making available valuable information for frequency stability.

1.1. State-of-the-art

There are examples of local inertia estimation, i.e. synchronous generator inertia estimation using synchrophasors [2–5]. These methods do not address the systematic estimation of inertia, considering the cautious time of disturbance detection since it varies as the disturbance propagates across the system. Some of these methods approach the disturbance time detection, like [3,5]. However, such ones are heuristic or depend on user interference. The inertia estimation methods in power grids can be classified according to the type of signals used for assessment. The methods may require pre-determined user injected signals to the grid, i.e. probing signal methods [6]. Typical nominal operation signals, i.e. ambient data methods [7–13]; or signals from severe events, like generation or line trips, i.e. rigorous disturbance methods [14–20].

Probing signal methods use a known and small perturbation to a signal of interest for inertia estimation, usually active power. The

* Corresponding author.
E-mail address: azamoram@umich.mx (A. Zamora-Mendez).

https://doi.org/10.1016/j.ijepes.2023.109323
Received 30 November 2022; Received in revised form 28 April 2023; Accepted 16 June 2023
Available online 1 July 2023
0142-0615/© 2023 Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

tiny signals are typically periodic, with known window length and magnitude. For example, a microperturbation method (MPM) is applied to a closed-loop generator which is assumed to be close to the centre of inertia (COI) of the system, and is used for estimation of the regional inertia [21]. The authors in [6] propose a regional inertia estimation method by regulating electrical storage systems (ESS) using periodic small load step perturbations. The inertia is estimated by the impulse response determined using the least-square method applied to the frequency measured at the ESS bus.

Ambient measurement methods utilise signals corresponding to the power system nominal operation. However, these signals are subject to a higher signal-to-noise ratio (SNR); thus, they impose more extensive and complex pre-processing stages. Likewise, these methods usually exhibit low accuracy levels. Ambient measurement methods can be further divided into tie-line monitoring methods [8]–[9], COI frequency representing regional equivalent machine [7,11–13], and artificial intelligence methods [10].

By employing the information of electromechanical oscillations (frequency, damping and mode shape) among areas, relations are derived from modal patterns and regional inertia. The method in [8] is validated using the IEEE 39-bus test system and laboratory experiments. In contrast, investigations in [9] are validated using the IEEE 68bus NETS-NYPS test system and actual data from the China northern system.

In [7], the estimation employs frequency measurements associated with generators and active power measurements corresponding to tie-lines. They are used as input and output measurements for ARMAX-based model estimation. This method considers the COI frequency as the equal-weighted mean of generator frequencies and is tested with actual measurements from the Icelandic power system. In [11], the estimation of regional inertia advocates a deterministic periodic portion and a noisy portion, depending on the region's renewable generation participation. The assessment is done using an autoregressive (AR) model and tested in an Italian system region. The method proposed in [12] utilises measurements from generation buses. The COI's frequency is estimated as the generators' frequencies weighted average, with such weights as the reciprocal of their variance. The estimation is done using the N4SID algorithm, and the sliding window used for updating the assessment has an exponential attenuation. This method is further investigated in [13], where the estimation considers the COI frequency approximated by the simple average of the generators' frequencies [7].

Another approach to ambient data measurements is developed through artificial intelligence tools. This approach is pursued in [10], where features are extracted from PMU signals, weather and load data. The estimation of the regional inertia is performed by a multivariate random forest regression (MRFR) algorithm, and the method is tested in the WECC system, with actual data recorded by the FNET/GridEye WAMS.

Severe disturbance methods use signals collected during significant events such as generation or line trips that stimulate the dynamic system response. Although these methods require circumstances that cannot be foreseen, they provide an accurate estimation. One strategy in this classification is reported in [22,23], where events are detected and selected using detrended fluctuation analysis (DFA). The inertia estimation deals with the ratio between the calculated power deviation of the dispatched generators and the measured WAMS response. This ratio is multiplied by the known inertia contribution of synchronous generators, and the surplus is attributed to load contributions. The authors in [14] propose the estimation of the effective inertia from disturbances measured at tie-lines among regions, where the active power deviation and the tie-line frequencies are used to estimate the inertia per region.

Regarding the artificial intelligence approach, regional inertia is estimated in [15] using particle swarm optimisation (PSO) applied to a cost function defined from the swing equation and considering the load contribution. The polynomial frequency response fitting is explored in [17], where the method requires the knowledge of the generators' inertia for approximating the COI frequency. In [16], the estimation of regional inertia seeks to match the generators' power response through the first-order nonlinear aggregated power system model in combination with the recently proposed dynamic regressor and mixing (DREM) procedure and the COI frequency response by the simple average of the generators' frequencies. Finally, a frequency domain approach is explored in [18]. The dynamic mode decomposition (DMD) technique extracts the eigenvalues and eigenvectors from disturbance data collected from WAMS, and the inertia is derived as a relation between it and the eigenvalues. The authors in [19] also use data from oscillation; however, the model fitting is performed using recursive least-squares (RLS), where the initial model is estimated using the non-recursive system identification method. Another method that explores frequency domain information collected from disturbance signals is found in [20], where the inertia of the Japanese 60 Hz system is estimated from mode shape analysis.

From the methods found in the literature, it is clear that the more accurate methods rely on severe disturbances, being regarded even as reference values for probing signal and ambient data methods. However, a few ways (e.g. [5,22,23]) discuss the importance of precisely estimating the moment of disturbance for appropriate inertia estimation. Furthermore, the swing equation directly links the inertia response to the rate-of-change-of-frequency (RoCoF). Therefore, it is especially relevant to identify the appropriate moment of disturbance, as more extensive errors in the derivative of frequency measurements will occur close to the disruption and must be appropriately processed. Despite the disturbance is detected by observing the time-varying properties of the extracted fluctuation, which is obtained using DFA. The authors in [22] claim that the detection is run over a sliding window with a length of 50 samples.

In contrast to some works, a new method of estimating the inertia in large systems using the relationship between the inertia support energy and the maximum frequency deviation is proposed in [24] without requiring the calculation of the COI frequency and the RoCoF.

Another approach is the inertia calculation application (ICA), which could be implemented as part of a Wide Area Monitoring Protection and Control scheme, is presented in [25]. The ICA aims to calculate the generator inertia available in individual locations of the system, to produce a profile of the inertia available in the system as a whole, immediately after the occurrence of a disturbance using a methodology based on the oscillation equation. In [26], a method is proposed for area inertia estimation of power system containing wind power considering the dispersion of frequency response and analysing the impact of the inertia distribution between areas on the frequency stability and inter-area oscillation.

Similarly in [27], a new methodology is proposed to estimate the transfer function parameters of synchronous generators (SG) and converter-interconnected generators (CIG), including the mechanical time constant and the primary frequency control (PFC) droop gain. The algorithm uses active power and bus frequency measurements from PMUs. Comparisons of different algorithms used for inertia estimation are summarised in Table 1. Furthermore, Refs. [28,29] provide a comprehensive summary of inertia definitions for both synchronous generators and CIGs as well as their corresponding estimation methods.

1.2. Contribution

Nowadays, the displacement of rotational generation in modern power systems has caused a reduction in system inertia, leading to a significant impact on the stability and reliability of power grids. Besides, the inertia reductions may cause significant RoCoF variations that may reach unacceptable levels after power imbalances, which might lead to large frequency deviations, cascading effects, or blackouts. In this context, this paper develops a novel methodology based on WAMS

Table 1
Comparison among different works for inertia estimation.

Ref.	Swing equation	Sliding window	Electromech. oscillations	System identif.	PMUs	Online	CIGs/ RES	SG	Regional system
[2]		X			X	X			X
[3]	X				X	X			X
[4]			X		X			X	
[5]	X			X	X	X		X	
[6]				X	X				X
[7]				X	X				X
[8]	X		X		X				X
[9]			X		X	X			X
[11]				X	X	X	X		X
[12]	X	X		X	X	X			X
[13]					X	X	X		X
[15]	X						X		X
[16]				X	X	X			X
[17]	X						X		X
[18]			X		X	X			X
[19]			X	X	X	X	X		X
[24]					X	X			X
[25]	X				X	X			X
[26]		X	X				X		X
[27]				X	X	X	X	X	
[30]	X		X		X				X
[31]	X		X						X
[32]	X			X			X		
[33]		X							
[34]			X		X				X
[35]				X	X	X			X
[36]					X	X		X	
[37]					X	X	X	X	
[38]					X	X			X
[39]						X		X	
Proposed		X			X	X		X	

information to estimate the inertia level in bulk power grids, including the most critical indicator of the robustness in power systems from the point of view of frequency stability, i.e., the RoCoF. Thus, the primary contributions of this paper are summarised as follows:

1. A model-free technique for estimating the inertia constant in bulk power grids derived from the second Newton law is presented.
2. The proposed methodology uses both active power deviation and frequency measurements stemming from WAMS, after load changes or generation trips, to estimate in a sole stage the differences in power and RoCoF.
3. The RoCoF is directly estimated by the first-order O-spline differentiator applied to the frequency measurements. At the same time-stamp, the amplitude of the active power deviation is calculated by the first O-spline lowpass filter used for the active power deviation measurements obtaining a noise-level reduction.
4. The inertia estimation is executed through the multivariate O-splines-based differentiators. After the disturbance time is detected via the Teager–Kaiser energy operator (TKEO), a sliding window approach is used, to process all time-synchronised signals in a sole stage.
5. Due to the O-splines derived from the discrete-time Taylor–Fourier transform (DTTFT), a high reduction rate of the computational burden is achieved. This makes our inertia estimation proposal suitable for their implementation in micro-controllers or digital signal processors with a reduced cost.

Thus, the proposed approach has the potential to precisely estimate the inertia constant in a short time and only utilises WAMS information, which will enable the system operators to deploy effective strategies to supplement the lost inertia and guarantee stability in bulk power grids.

2. Fundamentals

This section briefly describes the mathematical foundations for the proposed methodology to estimate the inertia constants in bulk power grids, taking advantage of the power system time-synchronised measured responses. First, the Teager–Kaiser energy operator and the O-spline fundamentals are described. Later, the RoCoF and inertia definitions from a mathematical perspective are also presented.

2.1. *l*th-order Teager–Kaiser Energy Operator

The disturbance point may be estimated via the TKEO. This nonlinear energy operator for continuous-time and discrete-time signals was developed by Teager [40] and subsequently introduced by Kaiser [41]. The TKEO is highly effective for numerous applications, including the development of the energy separation algorithm (ESA) for demodulating AM–FM signals, tracking speech modulations, and detecting events in non-stationary signals, where an event is evaluated under damped and undamped signals, signals with ambient noise, and real signals [42].

An essential part of the inertia constant determination is the detection time in which the event occurs (e.g. [3,5,22]). In this work, the Teager–Kaiser energy operator is adopted to detect better the disturbance instant in which the event takes place in the analysed signals. Thus, the following basic expression of the Teager–Kaiser energy operator of *l*th-order can be derived in discrete form for discrete-time signals as [43]:

$$\Psi_l[s_k(n)] = s_k(n)s_k(n+l-2) - s_k(n-1)s_k(n+l-1) \tag{1}$$

The second-order operator, $l = 2$, was first introduced by Kaiser based on Newton's law of motion applied to a mass model suspended by a spring of force constant. Namely, if $x(t) = Acos(\omega t + \phi)$ is the displacement produced by a mass–spring undamped linear oscillator with mass (m) and spring constant (k), then the total energy of this oscillator represented as the sum of the potential and kinetic energy is [44]:

$$\frac{1}{2}(kx^2 + m\dot{x}^2) = \frac{1}{2}mA^2\omega^2 \tag{2}$$

Therefore, the operator (Ψ) applied to $x(t)$ can track the energy (per half-unit mass) of the source that produced the oscillation signal $x(t)$ due to [41,44]

$$\Psi[A\cos(\omega t+\phi)] = A^2\omega^2 \tag{3}$$

where A is the amplitude of oscillation and ω stands for the frequency of the oscillation.

Thus, the discrete-time energy operator for the second-order to track the energy and detect the instant in which the event takes place is expressed by

$$\Psi_2[s_k(n)] = s_k^2(n) - s_k(n-1)s_k(n+1). \tag{4}$$

2.2. Estimation via O-splines

The FIR filter and its first-order differentiator used to estimate the amplitude of the active power deviation and the RoCoF, are respectively implemented with the O-splines of the DTTFT [45–47], expanding the Fourier subspace by incorporating higher-order (greater than zero) Taylor terms. These FIR filters attain the best Taylor–Fourier (TF) coefficients corresponding to the harmonic estimates at each frequency, allowing maximally-flat pass-band, and zero flat stop-band filters. For a given Taylor order K, the O-spline and its first K derivatives are orthogonal to the set of $K+1$ Taylor terms in the Taylor matrix, forming a bi-orthogonal pair of bases for the Taylor–Fourier subspace limited by that order. These filters reach the ideal bandpass filter (with maximally flat unit gain at its bandpass) as the order of the O-spline goes to infinity. In addition, the DTTFT can be factored into two separate matrices $\boldsymbol{Y}$ and $\boldsymbol{\Omega}$ (Taylor and Fourier operators, respectively) that can be applied consecutively to span the Taylor–Fourier subspace vector basis in $\boldsymbol{\Phi} = \boldsymbol{Y\Omega}$, where the dual matrix of $\boldsymbol{\Phi}$ is given by

$$\tilde{\boldsymbol{\Phi}} = \boldsymbol{\Phi}(\boldsymbol{\Phi}^H\boldsymbol{\Phi})^{-1} = \boldsymbol{Y}(\boldsymbol{Y}^H\boldsymbol{Y})^{-1}\frac{\boldsymbol{\Omega}}{N} = \tilde{\boldsymbol{Y}}\frac{\boldsymbol{\Omega}}{N} \tag{5}$$

where N is the number of samples per fundamental period and such that $\tilde{\boldsymbol{\Phi}}^H\boldsymbol{\Phi} = \boldsymbol{I}$. The above expression means that columns in $\tilde{\boldsymbol{\Phi}}$ are harmonic modulations of columns of diagonal submatrices in $\tilde{\boldsymbol{Y}}$.

Thus, a first Taylor order ($K=1$) O-spline is employed in this paper. The first lowpass O-spline and its first derivative are obtained in closed-form by solving the inverse matrix of $\boldsymbol{Y}$ by concatenating each one of the subvectors of $\tilde{\boldsymbol{\Phi}}$ [47]. Thus, for $K=1$, we have

$$\tilde{\boldsymbol{\Phi}}^{(1)} = \begin{pmatrix} 1 & t_1 \\ 1 & t_2 \end{pmatrix} \tag{6}$$

where the segments t_1 and t_2 are defined as $t_1 = t_{[-T,0)}$, $t_2 = t_{[0,T)} = t_1 + T$, and T denotes the fundamental period.

If $|\tilde{\boldsymbol{\Phi}}^{(1)}| = t_2 - t_1 = T$. Thus, we have [47]:

$$\tilde{\boldsymbol{\Phi}}^{(1)} = \frac{\begin{pmatrix} t_2 & -1 \\ -t_1 & 1 \end{pmatrix}}{T} = \begin{pmatrix} u_1+1 & -F_1 \\ -(u_2-1) & +F_1 \end{pmatrix} \tag{7}$$

where u_n is the normalised time: $u = t_n/T$, with $T = 1/F_1$, and F_1 is the bandwidth of the desired filter. Also, notice that the first column of $\tilde{\boldsymbol{\Phi}}^{(1)}$ is the triangular pulse given by [47]

$$\tilde{\varphi}^{(1)}(u) = \begin{cases} u+1 & \text{for} \quad -1 \le u < 0, \\ 1-u & \text{for} \quad 0 \le u < 1, \\ 0 & \text{otherwise}, \end{cases} \tag{8}$$

and that the second column corresponds to a Haar wavelet: $-F_1\dot{\tilde{\varphi}}^{(1)}(u)$.

The frequency responses of the first O-spline and its first differentiator are shown in Fig. 1. The top plot of Fig. 1 depicts the 0th differentiator (lowpass filter) frequency response, whereas the bottom one illustrates its first differentiator. Notice that they are centred at 0 Hz frequency (DC-component) and that both have zero-maximally-flat gain

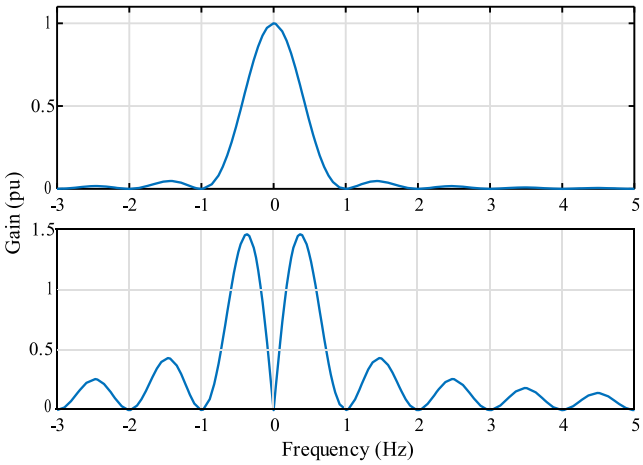


Fig. 1. Frequency responses of the O-spline (top plot) and its first differentiator (bottom plot) for $K=1$.

at the harmonic frequencies. This is an essential feature for both filters. The length of the FIR filters is given by

$$L = (K+1)N \tag{9}$$

where $N = F_s/F_1$, and F_s is the sampling frequency.

The main idea of implementing a DC-component filter is that the signals to be processed are non-sinusoidal signals, i.e., the active power deviation and frequency in each power system generator.

The phasors for multiple input signals are obtained using the best least mean squares (LMS) coefficients for projecting signals onto the Taylor–Fourier subspace. So, these are provided by

$$[\hat{\xi}_1 \quad \cdots \quad \hat{\xi}_q \quad \cdots \quad \hat{\xi}_m] = \tilde{\boldsymbol{\Phi}}^{(K)H}[\mathbf{s}_1 \quad \cdots \quad \mathbf{s}_q \quad \cdots \quad \mathbf{s}_m] \tag{10}$$

where the qth column $\mathbf{s}_q$ is denoted by the segment of a single signal, and vector $\hat{\boldsymbol{\xi}}_q$ contains the successive complex phasor derivatives corresponding for each $\mathbf{s}_q$ signal [45–47]:

$$\hat{\boldsymbol{\xi}}_q = [\hat{\xi}_q \quad \hat{\dot{\xi}}_q \quad \hat{\ddot{\xi}}_q \quad \cdots \quad \hat{\xi}_q^{(K)}]^T \tag{11}$$

In this work, $K=1$, and $\tilde{\boldsymbol{\Phi}}^{(1)}$ of (7) is implemented to compute the estimates by means of (10), so $\hat{\boldsymbol{\xi}}_q = [\hat{\xi}_q \quad \hat{\dot{\xi}}_q]^T$ is used to obtain the amplitude and its first derivative for each signal.

Then, the best instantaneous estimates of amplitude and its first derivative are given by [46]

$$\begin{aligned} &\hat{\boldsymbol{a}} = 2|\hat{\boldsymbol{\xi}}|; \qquad \hat{\boldsymbol{\varphi}} = \angle\hat{\boldsymbol{\xi}} \\ &\hat{\boldsymbol{a}} = 2\,\mathrm{Re}\{\hat{\boldsymbol{\xi}}e^{-j\hat{\boldsymbol{\varphi}}}\} \end{aligned} \tag{12}$$

where $\hat{\boldsymbol{a}} = [\hat{a}_1 \quad \cdots \quad \hat{a}_q \quad \cdots \quad \hat{a}_m]^T$, $\hat{\dot{\boldsymbol{a}}} = [\hat{\dot{a}}_1 \quad \cdots \quad \hat{\dot{a}}_q \quad \cdots \quad \hat{\dot{a}}_m]^T$, $\hat{\boldsymbol{\varphi}} = [\hat{\varphi}_1 \quad \cdots \quad \hat{\varphi}_q \quad \cdots \quad \hat{\varphi}_m]^T$, $\hat{\boldsymbol{\xi}} = [\hat{\xi}_1 \quad \cdots \quad \hat{\xi}_q \quad \cdots \quad \hat{\xi}_m]^T$, and $\hat{\dot{\boldsymbol{\xi}}} = [\hat{\dot{\xi}}_1 \quad \cdots \quad \hat{\dot{\xi}}_q \quad \cdots \quad \hat{\dot{\xi}}_m]^T$ respectively correspond to the amplitudes, first derivative of amplitudes, phase angles, complex phasor, and complex phasor first derivative for the set of analysed signals.

Given the data availability of frequency and active power deviation measurements stemming from PMUs assembled at each signal vector $\mathbf{s}_m$ in the right-hand side of (10), the amplitude of the active power deviations and RoCoF estimations are obtained via the O-splines using (10)–(12) as follows

$$\Delta\hat{\mathbf{P}}_e = 2|\hat{\boldsymbol{\xi}}| \tag{13}$$

$$\widehat{\mathbf{RoCoF}} = 2\,\mathrm{Re}\{\hat{\dot{\boldsymbol{\xi}}}\} \tag{14}$$

where the lowpass filter is used to estimate the amplitude of active power deviations, and its first-order differentiator is applied to the frequency measurements to estimate the RoCoF. Notice that the phase angle is zero, for which only the amplitude and its derivative are required.

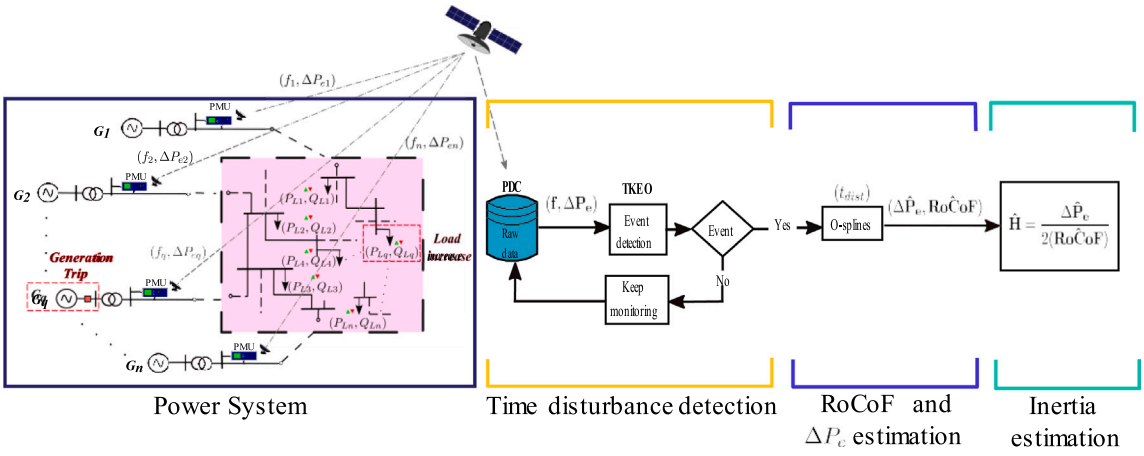


Fig. 2. Flowchart of the proposed methodology for inertia estimation in bulk power grids.

2.3. RoCoF and inertia

Inertia definition directly derives from the second Newton's law as:

$$F = m \cdot a = m \cdot \frac{dv}{dt} \tag{15}$$

where F is the resulting force (N), acting on a mass (kg) and producing an acceleration (m/s^2). For electric systems, this is referred to as a rotating mass with an angular velocity (Ω), which can be expressed as [48]:

$$T_m(t) - T_e(t) = J \cdot \frac{d\Omega}{dt} \tag{16}$$

where T_m corresponds to the mechanical torque in N m, T_e denotes the electromagnetic torque in N m, and J is the total moment of inertia of the system, expressed in kg m^2. If we indicate the total kinetic energy of rotating masses as

$$E_{kin} = \frac{1}{2} \cdot J \cdot \Omega^2 \tag{17}$$

The kinetic energy varies if a power imbalance is imposed on the system so that we can introduce a relationship between active power balance in the system and speed deviation as:

$$\Delta P_M(t) - \Delta P_E(t) \approx \Omega_r \cdot J \cdot \frac{d\Omega}{dt} \tag{18}$$

where Ω_r (377 rad/s) is the nominal value of Ω.

Since the equivalent inertia of the system can be expressed by the relationship between the kinetic energy of the generators' rotors and the rated power S (in MVA) of synchronous machines as:

$$H = \frac{J \cdot \Omega_r^2}{2 \cdot S} \tag{19}$$

where H is quantified in seconds. Considering (18), we can re-write (19) as

$$\Delta P_M(t) - \Delta P_E(t) = \frac{2SH}{\Omega_r} \cdot \frac{d\Omega}{dt} \tag{20}$$

or, alternatively, its representation in per unit becomes

$$\Delta P_m(t) - \Delta P_e(t) = 2 \cdot H \cdot \frac{d\omega}{dt} \tag{21}$$

Due to the mechanical power maintaining its steady-state or presenting only a tiny change in the inertial process [49], the inertia constant (in seconds) of the rotating machines can be represented as follows:

$$H = -\frac{\Delta P_e}{2 \cdot \frac{d\omega}{dt}} = -\frac{\Delta P_e}{2 \cdot RoCoF} \tag{22}$$

This paper processes ΔP_e and RoCoF estimates via the first O-spline and its derivative resulting from the solution of the DTTFT in Section 2.2. To assess the inertia constants in bulk power grids using measurements stemming from PMUs (frequency and active power), the expression in (22) can be extended as follows:

$$\hat{\mathbf{H}} = -\frac{\Delta \hat{\mathbf{P}}_e}{2 \cdot \hat{\mathbf{RoCoF}}} \tag{23}$$

where $\Delta \hat{\mathbf{P}}_e$ and $\hat{\mathbf{RoCoF}}$ respectively correspond to the estimations of the amplitude of the active power deviations and RoCoF via multivariate O-splines for all generation units using (13)–(14).

3. Proposed methodology for inertia estimation

The proposed methodology for inertia estimation in bulk power grids using measurements stemming from PMUs is illustrated in Fig. 2, where the following main items are addressed.

3.1. Detection of disturbance time

For this purpose, the TKEO is in charge to elucidate the right disturbance instant on the signals. First, the frequency signal energy is examined in (4) to find such time. This is conveyed by taking three continuous frequency samples to compute the instantaneous energy value (Ψ). These samples advance over the signal in a sliding way. If the energy value overpasses the zero value (steady-state condition), then the right disturbance instant (t_{dist}) is determined.

3.2. ROCOF and ΔP_e estimations

Once the disturbance time is detected, all generators' active power deviation and frequency signals are processed by means of the O-splines methodology. Thus, this section explains the RoCoF and ΔP_e estimations via O-splines. As proven in [47], the O-splines of the DTTFT provide FIR filters whose frequency response is similar to an ideal filter, which ensures smooth and accurate estimates. Due to this reason, this investigation adopts the O-splines estimates to process oscillating signals triggered by load increase and generation trip disturbances. These signals can be either generated by simulations or raw data from single or multiple IEDs. To process those signals (frequency and active power), the FIR filter and its first order differentiator are designed according to Section 2.2 with a sampling rate F_s = 1 kHz, K = 1, a bandwidth F_1 = 1 Hz, and a central frequency at 0 Hz (DC-component) as can be seen in Fig. 1. Since K = 1 is used, this implies the selection of a two-cycle time window of 2 seconds or 2000 samples, based on (9).

The RoCoF is derived as the outcome of the first differentiator in (14) when frequency signals are analysed. The first RoCoF estimate occurs after the first window is filled out and corresponds to the centre of the observation window. The best estimated RoCoF values are obtained

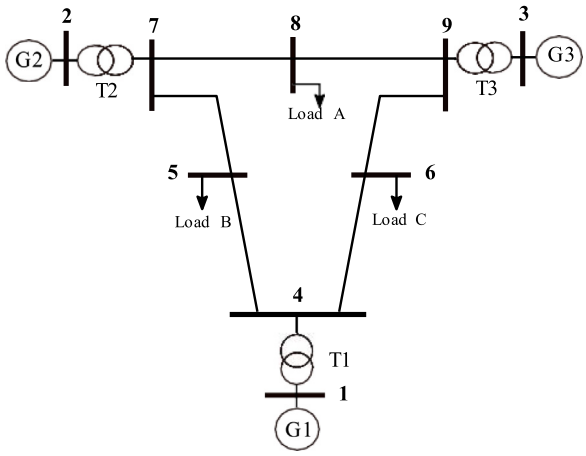


Fig. 3. One-line diagram of the WSCC system.

when the analysis window is over the slope of the analysed signal, i.e., those estimates just after the disturbance occurs. Afterwards, the obtained RoCoF values are used for the inertia estimation (23).

In the same way, to prevent large oscillations and inadequate values of inertia estimations, the O-splines also estimate the amplitude of the active power deviation signals in (23). This allows a smooth and accurate assessment of the amplitude of the active power deviation signals through the first O-spline lowpass filter using (13).

3.3. Inertia estimation

Finally, the inertia estimation in (23) is based on assessing the amplitude of the active power deviation and RoCoF via O-splines, considering a 2s sliding window just after the disturbance is detected. Then, the best estimate is found at the centre of the observation window, i.e., one second after the disturbance instant.

4. Benchmark bulk power grids for inertia estimation

This section simulates two well-known benchmark power grids in Power System Toolbox (PST) [50]. These are [51]: (*i*) a 3-machine and 9-bus system (S1) known as WSCC in Fig. 3; and (*ii*) a 16-machine and 68-bus system (S2) known as New England-New York in Fig. 4. All simulations consist of a total simulation time of 20 s, event inception at 2s, and a sampling frequency of 1 kHz. All test systems were assessed under load increasing scenarios and generation trips, deriving two cases per system, C1 and C2, respectively. The load increase case of (*C1*) for systems S1 and S2 is performed by the *lmod* and *rlmod* functions of PST to control the load changes in active and reactive power, respectively. Both functions perform modulations via load control equations employing the signal *lmod_sig*. Meanwhile, the generation trip case (*C2*) is conducted utilising the switching data in *sw_con* to define the type of fault. Each trip is implemented on the line that directly connects the generation to be tripped.

In any case, the reading of these functions is contained in *s_simu*; the *s_simu* script calls the PST models for data file selection, load flow development and solution, initialisation of the nonlinear simulation models, and step-by-step integration of the dynamic equations to respond to a user-specified system fault, following the procedure in [50].

For all cases, the error calculation is obtained with the relative error by

$$\%\mathbf{RE} = \frac{|\mathbf{H_{ref}} - \hat{\mathbf{H}}|}{|\mathbf{H_{ref}}|} \times 100 \tag{24}$$

where $\mathbf{H_{ref}}$ is a vector containing the constant inertia actual value for all analysed generators and $\hat{\mathbf{H}}$ denotes the inertia estimates obtained by the proposed methodology.

5. Results and validation for model-free inertia estimation

5.1. Western system coordinated council (WSCC)

This system is composed of 9-bus, 3-generator and three large equivalent loads connected in a meshed transmission network through transmission lines. The base MVA is 100, and the system frequency is 60 Hz.

5.1.1. Application on case S1.C1

This first case increases load B located at bus 5, as shown in Fig. 3. The load event occurs at $t = 2$ s, increasing the load by 40% concerning the active and reactive power demand at bus 5. The load changes from $P = 125$ MW and $Q = 20$ MVAR up to $P = 175$ MW and $Q = 70$ MVAR. The inception of this disturbance triggers the oscillating behaviour observed in the power deviation plot between 2s and 10 s, as exhibited in Fig. 5(a), where a new operating condition is established at the end of the simulation time, 10 s. In the meantime, the system experiments a large frequency excursion, dropping down from 60 Hz to 57 Hz, Fig. 5(c). Both power deviation and frequency excursions make up the input for the O-splines after the TKEO detects the disturbance.

Thus, after achieving the disturbance detection time (t_{dist}), a 2s rectangular sliding window is buffered over the frequency signals processed by the O-splines, as indicated in the grey dotted square in Fig. 5. Then, the RoCoF estimations are drawn in Fig. 5(d) according to (14). Similarly, the envelopes of the power deviations ($\Delta\mathbf{P}_e$) are extracted through the multivariate analysis performed by O-splines using (13), and they are depicted in Fig. 5(b).

Once the power deviations and RoCoF estimations are obtained via O-splines, the best estimates are in half of the window marked in the dotted line (i.e. one second after the disturbance). The inertia estimation can be determined through (23). Fig. 5(e) illustrates the inertia estimation compared with its respective reference, indicated by a grey dotted line. The results show the continuous behaviour of the inertia estimate as a function of time for all generators after the generated event. Remarkably, the inertia estimation is quite precise, guaranteed by the formulation established in Section 2. Thereby, the actual values of the inertia constant of all generators and the estimated inertia via O-splines are practically the same.

5.1.2. Application on case S1.C2

The second case simulates a generation trip on generator G3 of the system shown in Fig. 3. This trip causes a generation loss of 85 MVA, i.e., 26.6% of the power generated by the system. The event is generated at the instant when $t = 2$ s, and the signals for this case are shown in Fig. 6.

Once the disturbance is applied to the system, an oscillatory response is unleashed through the generators' electrical power deviation, with a decrease in frequency below 55 Hz, Figs. 6(a) and (c), respectively. After emerging these variations and according with the proposed methodology, the event detection process is activated to obtain the instant $t_{dist} = 2s$, as shown in Fig. 6(a). Subsequently, the analysis via O-splines is applied to the power deviation and frequency signals, starting from the point where the event is detected and results in the estimation of power deviation ($\Delta\hat{P}_e$) and $\hat{RoCoF}$, as exhibited in Fig. 6(b) and (d). From the attained results, it is relevant to point out that the first estimate also coincides with the centre of the window, which is marked using the grey dotted square in all plots of Fig. 6(b) and (d)–(e).

The final stage of the methodology consists of the inertia constants' estimation through (23) by employing the power deviation and RoCoF estimates, whose results are depicted in Fig. 6(e). The inertia remains constant after the event, except for generator G3, which goes out of operation. They are compared with the reference inertia for each generator by using a grey dotted line.

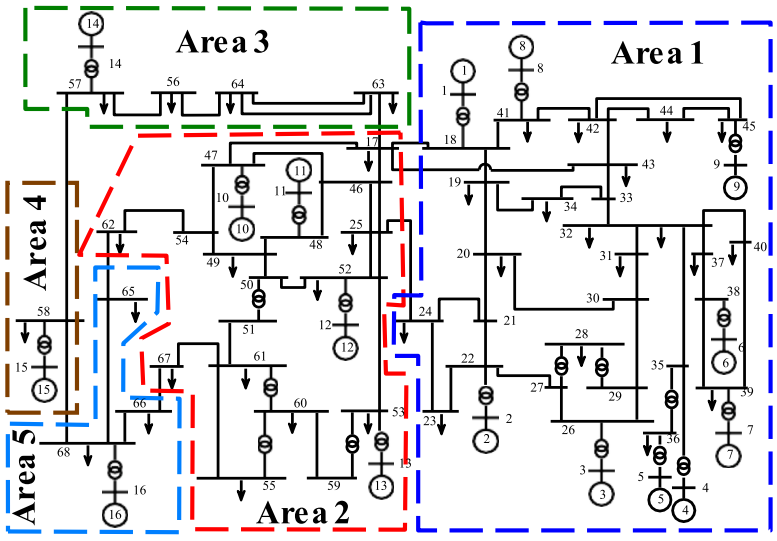


Fig. 4. Single-line diagram of NETS-NYPS.

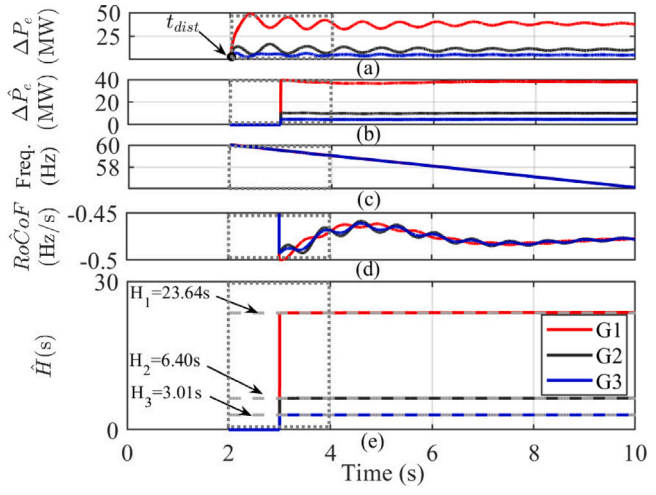


Fig. 5. *Case S1.C1*: RoCoF and inertia estimations to a 40% load increase at bus 5 of the WSCC system.

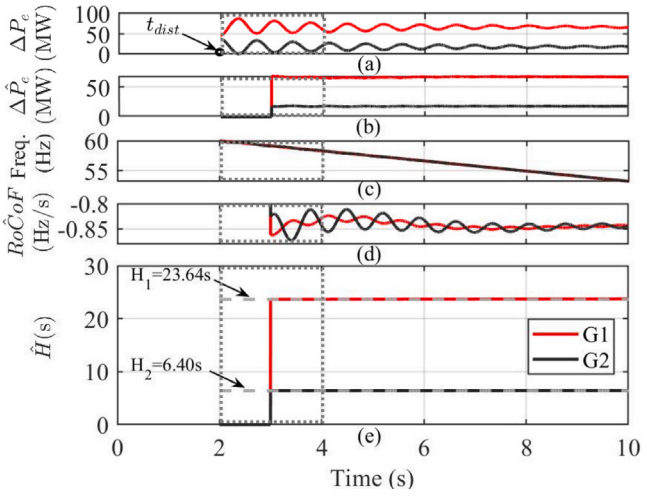


Fig. 6. *Case S1.C2*: RoCoF and inertia estimations after tripping generator G3 in the WSCC system.

To quantitatively describe the attained results, they are summarised in Table 2, where all generators and their inertia values are presented in the first two columns, respectively. For the *S1.C1* case, the average RoCoF values corresponding to a 2s window after the first estimation, the inertia constants and their relative errors (RE) concerning the reference values in (24) are listed in columns 3–5 of Table 2, respectively. Similarly, for the *S1.C2* case, the estimated values and errors are presented in columns 6–8. Note that the attained values for the relative error for both study cases in the WSCC system are less than 0.05%.

Since the inertia constants can be represented as the sum of all inertial constants associated with each generator, the last row in Table 2 shows this total inertia of the system ($\sum H_i$) and its relative error for the actual value. Notice that the relative error **RE** obtained for the proposed method for inertia estimation is also lower than 0.05%, bringing high accuracy in the assessment.

For comparison purposes, the inertia calculation application (ICA) method in [25] is employed and its results are depicted in the last two columns of Table 2. The power system inertia calculation (H_{sys}) in the 3-machine power system is computed for both strategies and presented in the last row of Table 2. Notice that (H_{sys}) obtained by the ICA method is also presented in Table 2 of [25] for the sudden generator disconnection disturbance. Then, the comparison is only made disconnecting generator G3, where a power system inertia of 9.39125 s is estimated by the ICA method for an MVA base of S_B = 320.2 MW. Thus, a change of base is applied to compare with our proposal, using a new base of S_B = 100 MVA, resulting in H_{sys} = 30.0708 s. Besides, a minimum relative error of 1.39% is computed using the ICA method [25], meanwhile with our proposal a relative error of 0.03% is obtained for both test cases, being less than the reported in [25], demonstrating the effectiveness and outperformance of the O-splines inertia estimation method.

5.2. New England and New York interconnected system (NETS-NYPS)

This interconnection is very well-known by its historical record of oscillating behaviour. It comprises five interconnected areas that represent a reduced model of the New England interconnected system (Area 1), the electric power network of New York (Area 2), and the interconnections with Ontario Hydro, MISO and PJM power networks are represented by Area 3 to Area 5, respectively, as shown in Fig. 4. Each generator is modelled by a fourth-order model equipped with its excitation system.

Table 2
WSCC system inertia estimation via O-splines.

Gen	H_{ref} (s)	O-splines			O-splines			ICA method [25]	
		Load increase			Generation trip			Generation trip	
		$\widehat{RoCoF}$ (Hz/s)	$\hat{H}$ (s)	RE (%)	$\widehat{RoCoF}$ (Hz/s)	$\hat{H}$ (s)	RE (%)	$\hat{H}$ (s)	RE (%)
G1	23.64	−0.47	23.65	0.05	−0.83	23.65	0.05	23.97	1.39
G2	6.40	−0.47	6.40	0.04	−0.83	6.40	0.04	6.10	4.67
G3	3.01	−0.47	3.01	0.03	8.46	–	–	3.42	13.49
$H_{sys} = \sum H_i$	33.05	–	33.06	0.03	–	30.05	0.03	33.49	1.32

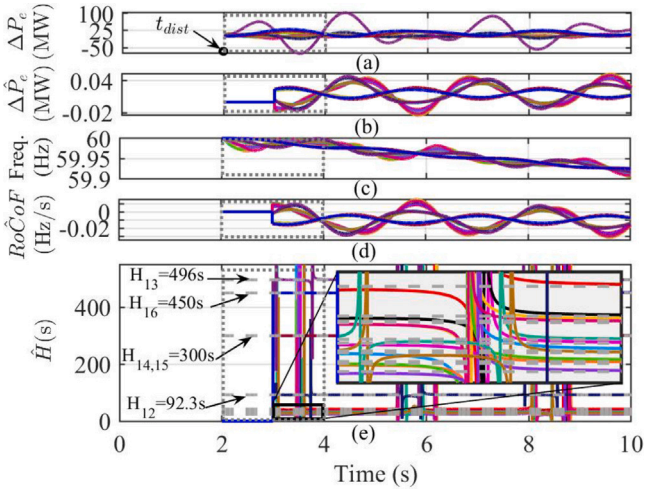


Fig. 7. *Case S2.C1*: RoCoF and inertia estimations to a 40% load increase at bus 32 of the NETS-NYPS system.

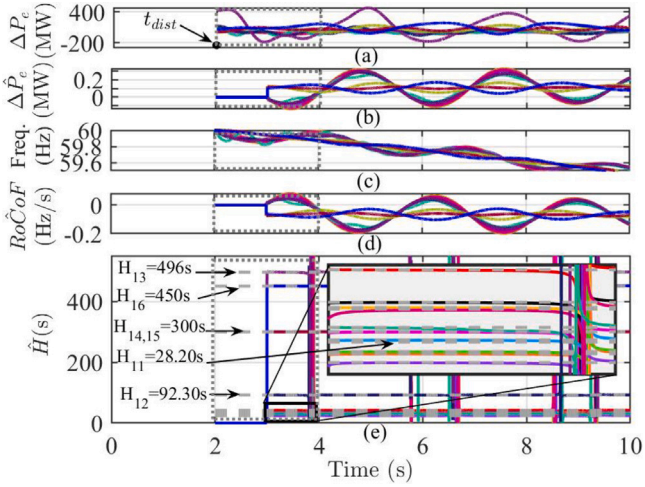


Fig. 8. *Case S2.C2*: RoCoF and inertia estimations after tripping generator G11 in the NETS-NYPS system.

5.2.1. Application on case S2.C1

This case implements a load increase at bus 32 (Medway substation) in Fig. 4. This event produces a load increase of 40% to the substation's rated demand (real and reactive powers). The load changes from P = 329 MW and Q = 32.3 MVAR up to P = 460.6 MW and Q = 45.22 MVAR. As a consequence, power deviations and frequency signals in Figs. 7(a) and (c) fluctuate around the operating point during the simulation, allowing us to run our methodology in Section 3 on a large-scale power grid. Thus, the TKEO strategy is conducted to elucidate the right disturbance instant at t = 2 s, setting a 2s analysis window to carry out the estimations via O-splines of the ΔP_e amplitude ($\Delta\hat{P}_e$) and RoCoF, as results in Figs. 7(b) and (d), respectively. Then, such estimations compute the inertial constant (23) for all generators at every instant, as depicted in Fig. 7(e), where they are also compared against the actual values (grey dotted lines). Note that the time derivatives that provide the RoCoF estimations in Fig. 7(d) track the frequency changes observed in Fig. 7(c), where frequency fluctuations are also mimicked in the $\widehat{RoCoF}$ estimations. This fact impacts the inertia estimation generating high peaks, since RoCoF estimations near zero (zero crossings in Fig. 7(d)) are produced.

5.2.2. Application on case S2.C2

In this case, a generator trip is implemented on G11 in Fig. 4. This event causes a generation loss of 1000 MW, triggering an oscillatory response visible in both power deviations and frequency signals in Figs. 8(a) and (c), respectively. Notably, a frequency drop below 59.6 Hz is exhibited to apply our proposed methodology accordingly.

In this case, a disturbance time (t_{dist}) is also detected at t = 2 s by the TKEO, delimiting the multivariate O-splines analysis to the detected position and establishing the starting point of the analysis window to process the input signals. Then, estimations of $\Delta\hat{P}_e$ and $\widehat{RoCoF}$ are yielded in Figs. 8(b) and (d). Finally, a vector of inertia estimations $\hat{\mathbf{H}}$ is drawn by computing both assessments in (23), as displayed in Fig. 8(e).

To compare our methodology quantitatively against reference values, Table 3 lists the inertia estimations for all generators at each event above. It should be noted that all inertia values result from averaging over one second from the first estimation of H. This is done to solve the peaks generated during the analysis due to the $\widehat{RoCoF}$ zero crossings. In Table 3, inertia estimations are similar to reference values. They exhibit relative errors below 1.9% under the load increase case and ostensibly reduced below 0.7% in the generation trip case. After averaging, all global relative errors result in less than 0.1% and 0.05% for cases *S2.C1* and *S2.C2*.

Besides validating the proposed methodology against reference values, ARMAX-based state-of-the-art algorithm in [5] is also compared against our proposal, resulting in columns 7–8 in Table 3. Notice that the ARMAX-based inertia estimation method is not multivariate. Then, this methodology requires performing many disturbances as inertia constants need to be estimated. Nevertheless, it exhibits its highest error of 3.07% and a global relative error of 0.89%. Thus, the O-splines-based inertia estimation outperforms the ARMAX-based strategy. Finally, note that the results obtained with our proposal minimise the error in the analysis proposed in [5]; this demonstrates the effectiveness of the assessment via O-splines applied to both small and robust systems and reduces the computational cost involved in this type of multi-signal processing.

5.2.3. Application on case S2.C2 with penetration of wind power plants (S2.C2w)

The generator trip in *Case S2.C2* is now implemented into a scenario with penetration of wind power plants (WPPs), termed *S2.C2w*, allowing to extend the proposed method to the case when CIGs are included. Four WPPs are added, similarly as in [52]. Thus, the NETS-NYPS grid in Fig. 4 was modified by adding 4 transmission lines and creating 4 new generation buses, these lines interconnect each WPP with existing buses

Table 3
Inertia estimations in the NETS-NYPS system via O-splines and comparison against reference values of the ARMAX-based inertia estimation method [5].

Gen	H_{ref} (s)	O-splines Load increase		O-splines Generation trip		ARMAX [5] Generation trip		O-splines Generation trip in case *S2.C2w*	
		$\hat{H}$ (s)	RE (%)	$\hat{H}$ (s)	RE (%)	$\hat{H}$ (s)	RE (%)	$\hat{H}$ (s)	RE (%)
1	42.00	42.42	1.00	41.93	0.16	41.58	1.00	42.4746	1.1301
2	30.20	29.86	1.14	30.13	0.23	30.32	0.41	30.5480	1.1523
3	35.80	35.40	1.12	35.74	0.16	35.62	0.49	36.1853	1.0762
4	28.60	29.14	1.89	28.54	0.20	28.43	0.59	28.9970	1.3881
5	26.00	26.35	1.34	25.94	0.22	26.29	1.12	26.4127	1.5874
6	34.80	35.36	1.61	34.75	0.14	35.13	1.83	35.2460	1.2816
7	26.40	26.87	1.77	26.34	0.23	26.71	1.16	26.7695	1.3994
8	24.30	24.55	1.01	24.22	0.32	24.31	0.05	24.7268	1.7562
9	34.50	34.85	1.00	34.26	0.68	34.32	0.53	35.2563	2.1921
10	31.00	31.14	0.46	30.79	0.69	30.84	0.54	31.6032	1.9457
11	28.20	27.69	1.82	–	–	–	–	–	–
12	92.30	92.22	0.08	92.23	0.07	90.52	1.93	92.9274	0.6798
13	496.00	496.04	0.01	495.87	0.03	489.59	1.29	499.5860	0.7230
14	300.00	300.01	0.00	299.97	0.01	290.79	3.07	300.0860	0.0287
15	300.00	299.96	0.01	300.03	0.01	296.21	1.26	300.0607	0.0202
16	450.00	449.96	0.01	450.19	0.04	453.93	0.87	450.0675	0.0150
$\sum H_i$	1980.10	1981.81	0.09	1979.22	0.05	1963.37	0.89	1989.2	0.46

(17, 37, 43, 60). All WPPs are equipped with type-3 wind generators consisting of doubly-fed induction generators equivalent models, with 30% of power output injected via inverters with voltage regulation and unitary power factor with a dispatch of 100MW each plant.

After tripping the generator, the whole system evidences an oscillatory response in the power deviation signals and a frequency drop of 6dHz, as displayed in Figs. 9(a)–(c), respectively. Then, the disturbance instant is properly captured by the TKEO-based detection stage at $t_{dist} = 2s$, activating the second stage based on the O-splines technique which is simultaneously applied to the power deviation and frequency signals, and being able to estimate their estimates $\Delta\hat{P}_e$ and $\hat{RoCoF}$, as exhibited in Fig. 9(b) and (d). Where the analysis window is represented using the grey dotted square in all plots of Fig. 9(a)–(e), and the first inertia estimates $\hat{H}_i$ are located at its centre. All inertia constants are computed for all the remaining generators as depicted in column 9 in Table 3, together with their REs in column 10. It is noticed that despite the presence of WPPs degrading the inertia estimation, the relative errors only reach up to 2.9% for generator G9 and 0.46% in the total system inertia, showing high accuracy in the assessment.

5.3. Application on real data

The proposed methodology is also demonstrated by processing data associated with a real disturbance recorded by a PMU of the NYPS on December 7th of 2017, at 19:56:18 Eastern Time, which is employed in this research for inertia estimation purposes [5]. During this event, the PMU measured positive-sequence voltages and currents, sampled at 30 Hz, close to a power plant with a rated power of 849.1 MVA. The steam generator is from a nuclear power plant, which supplies base load and must operate close to its capacity continuously. The reference value for the inertia constant of this power plant at a 100 MVA system base is 36.51 s. This event is caused by a generation trip in the North American Eastern Interconnection and is labelled as GT_1 in [5]. Active power and frequency measurements for this disturbance are illustrated in Fig. 10. The tripping time of the disturbance is 302.5s. Notice that the frequency decreases from 60 Hz to 59.945 Hz in 310.4333 s, as can be observed in the top plot of Fig. 10 and in its zoomed area; meanwhile, the active power has an initial step of 10 MW which is showed in the zoomed area of the bottom plot of Fig. 10 and decays about 737.4534 MW.

To demonstrate the effectiveness of the methodology both frequency and real power excursions feed the O-splines-based methodology to estimate the inertia constant. Both signals are processed by a sliding window using the O-splines. Once the TKEO strategy identified the right disturbance instant at $t_{dist} = 302.9667s$, the O-splines's estimates are yielded in accordance with Section 2.2 and are denoted by the orange line in Figs. 11(a)–(b), with their corresponding raw data in blue colour. It is noteworthy to highlight the smoothness of the estimates, given that no pre-processing is required. Since our proposal can provide time derivatives, the RoCoF estimation is available exhibiting two zero crossings in Fig. 11(c) at $t = 307.6$ s and $t = 308.1$ s, in turn, causing the peaks in the inertia estimation due to division by zero in (23), see Fig. 11(d). For analysis purposes in the time inertia estimation, this

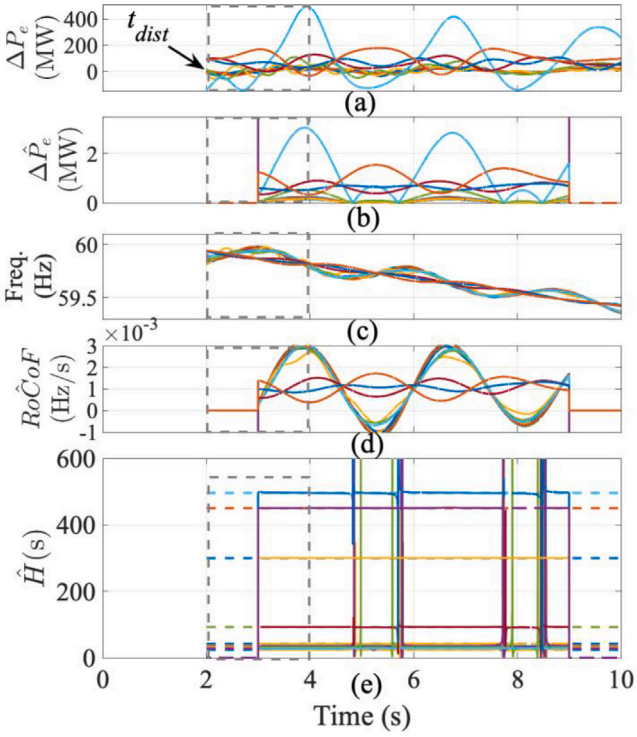


Fig. 9. *Case S2.C2w*: RoCoF and inertia estimations after tripping generator G11 in the NETS-NYPS system with penetration of wind power plants.

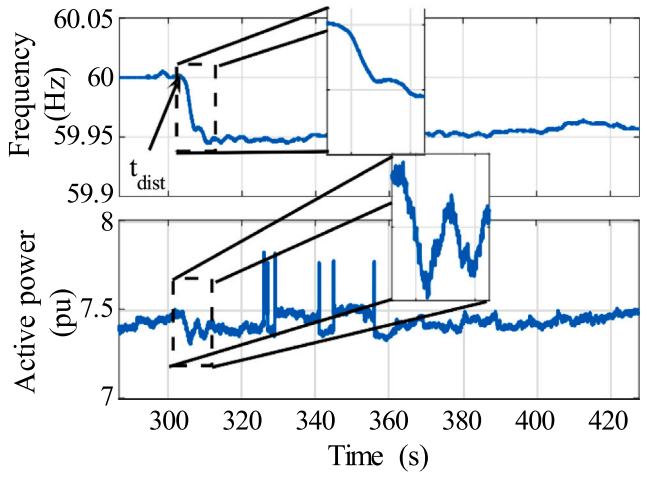


Fig. 10. Frequency and active power measurements captured by a PMU of the New York power system.

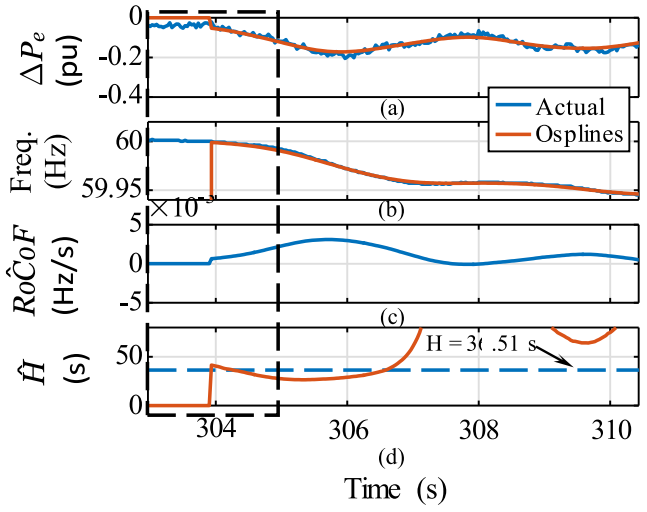


Fig. 11. Inertia estimation after trip generation in the North American Eastern Interconnection. (a) Frequency and its estimate, (b) power deviation and its estimate, (c) RoCoF estimate, and (d) inertia estimation.

results in an average inertia estimate of 35.9437s for the first three seconds. The dashed line box in Fig. 11 indicates the first analysis window with a length of 60 samples (K = 1) that is equal to 2 cycles of 1 Hz (bandwidth frequency). Thus, with the O-splines methodology, the relative error is $RE = 1.551\%$ being less than the reported in Table 4 for the GT_1 case in [5], where a RE of 5.64% is reported.

6. Discussion

6.1. Method performance in case of unobservability or missing PMUs

Actual WAMS can face observability problems or missing PMUs at some substations, then our methodology can also be fed by using power and frequency signals coming from dynamic state estimation stages or adjacent buses. Thus, when there is no complete observability of the system or simply no observability at all, this is not a limitation for this work.

To show the methodology performance when all generators are not equipped with PMUs, the inertia estimation of generator G3 in the WSCC system is considered, taking the load increment case as explained in Section 5.1.1. In this case, the required data are voltage phasor measurements and active power flows at adjacent buses to the G3. Due to the power and voltage measurements at the generator G3 are not available, then we use the active power flowing from bus 9 to bus 8 and from bus 9 to bus 6, see Fig. 3, and the frequency is obtained as the derivative of the voltage phasor angle at bus 9. It is worth mentioning that the losses attributed to the line are not considered in this case. Thus, if the power flow and frequency measurements are already available at bus 9, the generator inertia estimation is done in the same way as in the previous cases, resulting in an inertia value of 3.02 s, with a relative error of 0.36% with respect to the reference value.

Table 4
Parameters for filter designing via the O-splines of the DTTFT varying the window length.

Filter	F_s(Hz)	F_1 (Hz)	K	$L(s)$
1	1000	4	3	1
2		2	2	1.5
3		1	1	2
4		1	2	3

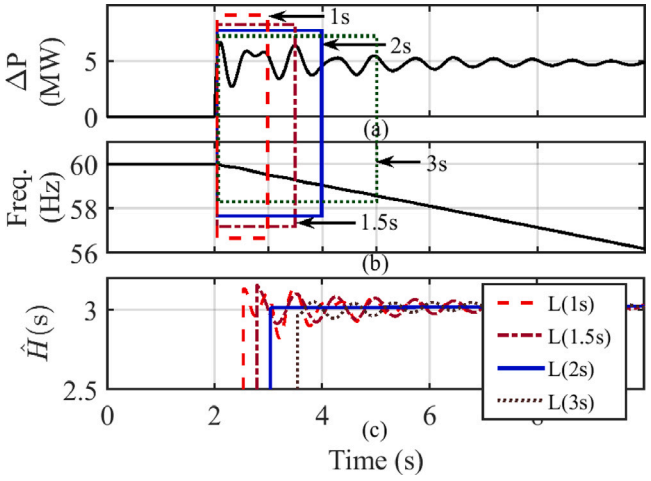


Fig. 12. Length variation in the sliding window for inertia estimation in the G3 generator of the WSCC system.

6.2. Window length analysis

The window length is defined through (9). For analysis purposes, this is varied by increasing the Taylor's order (K) and the bandwidth (F_1). Specifically, please also see Table 4. This table presents the variation of the window length using four FIR filters for bandwidth of 1, 2 and 4 Hz and Taylor's order of 1, 2, and 3; notice that the same sampling frequency is used for all cases. Thus, four inertia estimations of generator G3 in the WSCC are depicted in the bottom plot of Fig. 12, where it is remarkable that the appropriate window length is 2s.

Figs. 12(a)–(b) display all the window lengths considered for this analysis, exhibiting ΔP and frequency signals of the generator G3 in the WSCC system, Fig. 3. In Fig. 12(c), the inertia estimation is rendered for each window length, noting that the best estimation is for a window length of 2 s.

7. Conclusions

This paper has successfully demonstrated that our model-free strategy based on WAMS information can precisely estimate the inertia constant for each synchronous generator in bulk power grids. The proposed methodology uses active power deviation and frequency measurements from WAMS after load changes or generation trips. The

strategy's principal idea involves applying a DC-component filter designed by the first O-spline lowpass filter and its first corresponding differentiator to process multiple channels simultaneously. In this way, once the TKEO detects the disturbance time, the RoCoF and the amplitude of active power deviations are directly estimated by that pair of filters. Then, these are applied to the frequency and active power deviation measurements. Thus, this allows estimating the inertia constants derived from the second Newton's law in a sliding window approach, where all time-synchronised signals are quickly processed in a sole stage.

Moreover, since the high reduction rate of the computational burden is achieved with the DTTFT, this makes our inertia estimation strategy suitable for their implementation in micro-controllers or digital signal processors with a reduced cost. Furthermore, the proposed strategy was tested by using simulated signals of two well-known benchmark power grids simulated in the Power System Toolbox, the WSCC and New England-New York systems. Thus, the proposed approach supported by the O-splines technique combined with the TKEO technique is a powerful tool for inertia estimation in bulk power grids. Moreover, it exhibits reliable results with a global relative error less than 0.1% for all presented cases.

CRediT authorship contribution statement

Diego Rodales: Software, Investigation, Visualization, Data curation, Writing – original draft. **Alejandro Zamora-Mendez:** Conceptualization, Methodology, Formal analysis, Writing – original draft, Supervision, Investigation, Software. **José Antonio de la O Serna:** Supervision, Writing – review & editing. **Juan M. Ramirez:** Supervision, Writing – review & editing. **Mario R. Arrieta Paternina:** Conceptualization, Methodology, Formal analysis, Writing – original draft, Supervision, Investigation, Software. **Lucas Lugnani:** Investigation, Visualization, Writing – original draft. **Gabriel E. Mejia-Ruiz:** Writing – review & editing, Visualization. **Alexander Sanchez-Ocampo:** Writing – review & editing, Visualization. **Daniel Dotta:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

Authors gratefully acknowledge financial support from the Project Support Program for Research and Technological Innovation of UNAM (DGAPA, PAPIIT-2023-2025) through the project IT102723.

References

[1] Milano F, et al. Foundations and challenges of low-inertia systems. In: 2018 PS computation conf.. IEEE; 2018, p. 1–25.
[2] Wall P, Gonzalez-Longatt F, Terzija V. Estimation of generator inertia available during a disturbance. In: Power and energy society general meeting. Citeseer; 2012.
[3] Wall P, Terzija V. Simultaneous estimation of the time of disturbance and inertia in power systems. IEEE Trans Power Deliv 2014;29(4).
[4] Gorbunov A, Dymarsky A, Bialek J. Estimation of parameters of a dynamic generator model from modal PMU measurements. IEEE Trans Power Syst 2019;35(1):53–62.
[5] Lugnani L, Dotta D, Lackner C, Chow J. ARMAX-based method for inertial constant estimation of generation units using synchrophasors. Electric PS Res 2020;180:106097.
[6] Tamrakar U, Guruwacharya N, Bhujel N, Wilches-Bernal F, Hansen TM, Tonkoski R. Inertia estimation in power systems using energy storage and system identification techniques. In: 2020 international symposium on power electronics, electrical drives, automation and motion. IEEE; 2020, p. 577–82.
[7] Tuttelberg K, Kilter J, Wilson DH, Uhlen K. Estimation of power system inertia from Ambient Wide Area measurements. IEEE Trans PS 2018.
[8] Cai G, Wang B, Yang D, Sun Z, Wang L. Inertia estimation based on observed electromechanical oscillation response for power systems. IEEE Trans Power Syst 2019;34(6):4291–9. http://dx.doi.org/10.1109/TPWRS.2019.2914356.
[9] Wang B, Yang D, Cai G, Ma J, Chen Z, Wang L. Online inertia estimation using electromechanical oscillation modal extracted from synchronized ambient data. J Mod Power Syst Clean Energy 2022;10(1):241–4. http://dx.doi.org/10.35833/MPCE.2020.000105.
[10] Cui Y, You S, Liu Y. Ambient synchrophasor measurement based system inertia estimation. In: 2020 IEEE power & energy society general meeting. IEEE; 2020, p. 1–5.
[11] Allella F, Chiodo E, Giannuzzi GM, Lauria D, Mottola F. On-line estimation assessment of power systems inertia with high penetration of renewable generation. IEEE Access 2020;8:62689–97.
[12] Zeng F, Zhang J, Chen G, Wu Z, Huang S, Liang Y. Online estimation of power system inertia constant under normal operating conditions. IEEE Access 2020;8:101426–36.
[13] Baruzzi V, Lodi M, Oliveri A, Storace M. Analysis and improvement of an algorithm for the online inertia estimation in power grids with RES. In: 2021 IEEE international symposium on circuits and systems. IEEE; 2021, p. 1–5.
[14] Wilson D, Yu J, Al-Ashwal N, Heimisson B, Terzija V. Measuring effective area inertia to determine fast-acting frequency response requirements. Int J Electr Power Energy Syst 2019;113:1–8.
[15] Zografos D, Ghandhari M, Paridari K. Estimation of power system inertia using particle swarm optimization. In: Intelligent system application to ps (ISAP), 2017 19th int. conf. on. IEEE; 2017.
[16] Schiffer J, Aristidou P, Ortega R. Online estimation of power system inertia using dynamic regressor extension and mixing. IEEE Trans Power Syst 2019;34(6):4993–5001.
[17] Phurailatpam C, Rather ZH, Bahrani B, Doolla S. Measurement-based estimation of inertia in ac microgrids. IEEE Trans Sustain Energy 2019;11(3):1975–84.
[18] Yang D, Wang B, Cai G, Chen Z, Ma J, Sun Z, et al. Data-driven estimation of inertia for multi-area interconnected power systems using dynamic mode decomposition. IEEE Trans Ind Inf 2020.
[19] Makolo P, Zamora R, Lie T-T. Online inertia estimation for power systems with high penetration of RES using recursive parameters estimation. IET Renew Power Gener 2021.
[20] Kerdphol T, Watanabe M, Nishikawa R, Tamaki T, Mitani Y. Determining inertia of 60 Hz Japan power system using PMUs from power loss event. In: 2021 IEEE texas power and energy conference. IEEE; 2021, p. 1–5.
[21] Zhang J, Xu H. Online identification of power system equivalent inertia constant. IEEE Trans Ind Electron 2017;64(10).
[22] Ashton PM, Saunders CS, Taylor GA, Carter AM, Bradley ME. Inertia estimation of the GB power system using synchrophasor measurements. IEEE Trans PS 2015;30(2).
[23] Ashton PM, Taylor GA, Irving MR, Pisica I, Carter AM, Bradley ME. Novel application of detrended fluctuation analysis for state estimation using synchrophasor measurements. IEEE Trans Power Syst 2013;28(2):1930–8. http://dx.doi.org/10.1109/TPWRS.2013.2248027.
[24] Wang B, Sun H, Li W, Yang C, Wei W, Zhao B, et al. Power system inertia estimation method based on maximum frequency deviation. IET Renew Power Gener 2022;16(3):622–33. http://dx.doi.org/10.1049/rpg2.12367.
[25] Sun M, Feng Y, Wall P, Azizi S, Yu J, Terzija V. On-line power system inertia calculation using wide area measurements. Int J Electr Power Energy Syst 2019;109:325–31.
[26] Li D, Dong N, Yao Y, Xu B, Gao DW. Area inertia estimation of power system containing wind power considering dispersion of frequency response based on measured area frequency. IET Gener Transm Distribution 2022;16(22):4640–51. http://dx.doi.org/10.1049/gtd2.12628.
[27] Gotti D, Ledesma P, Amaris H. A Gray-Box System Identification Approach for Online Inertia Estimation Applications. 2022, http://dx.doi.org/10.36227/techrxiv.20495283.v1.
[28] Tan B, Zhao J, Netto M, Krishnan V, Terzija V, Zhang Y. Power system inertia estimation: Review of methods and the impacts of converter-interfaced generations. Int J Electr Power Energy Syst 2022;134:107362. http://dx.doi.org/10.1016/j.ijepes.2021.107362.
[29] Prabhakar K, Jain SK, Padhy PK. Inertia estimation in modern power system: A comprehensive review. Electr Power Syst Res 2022;211:108222. http://dx.doi.org/10.1016/j.epsr.2022.108222.
[30] Zhang Y, Qi F, He Y, Wang B, Yang D. Synchrophasor data-based inertia estimation for regional grids in interconnected power systems. Front Energy Res 2022;10. http://dx.doi.org/10.3389/fenrg.2022.989430.
[31] Wang B, Yang D, Cai G, Chen Z, Ma J. An improved electromechanical oscillation-based inertia estimation method. IEEE Trans Power Syst 2022;37(3):2479–82. http://dx.doi.org/10.1109/TPWRS.2022.3156441.

[32] Phurailatpam C, Rather ZH, Bahrani B, Doolla S. Estimation of non-synchronous inertia in AC microgrids. IEEE Trans Sustain Energy 2021;12(4):1903–14. http://dx.doi.org/10.1109/TSTE.2021.3070678.

[33] Yin H, Wu Y, Qiu W, Zeng C, You S, Tan J, et al. Precise ROCOF estimation algorithm for low inertia power grids. Electr Power Syst Res 2022;209:107968. http://dx.doi.org/10.1016/j.epsr.2022.107968.

[34] Kerdphol T, Watanabe M, Nishikawa R, Hayashi Y, Mitani Y. Inertia estimation of the 60 Hz Japanese power system from synchrophasor measurements. IEEE Trans Power Syst 2022;1. http://dx.doi.org/10.1109/TPWRS.2022.3168037.

[35] Liu J, Wang C, Bi T, Xu G. Online estimation of POI-level aggregated inertia considering frequency spatial correlation. IEEE Trans Power Syst 2022;1–13. http://dx.doi.org/10.1109/TPWRS.2022.3197129.

[36] Tan B, Zhao J, Terzija V, Zhang Y. Decentralized data-driven estimation of generator rotor speed and inertia constant based on adaptive unscented Kalman filter. Int J Electr Power Energy Syst 2022;137:107853. http://dx.doi.org/10.1016/j.ijepes.2021.107853.

[37] Guo J, Wang X, Ooi B-T. Estimation of inertia for synchronous and non-synchronous generators based on ambient measurements. IEEE Trans Power Syst 2022;37(5):3747–57. http://dx.doi.org/10.1109/TPWRS.2021.3134818.

[38] Guo J, Wang X, Ooi B-T. Online purely data-driven estimation of inertia and center-of-inertia frequency for power systems with VSC-interfaced energy sources. Int J Electr Power Energy Syst 2022;137:107643. http://dx.doi.org/10.1016/j.ijepes.2021.107643.

[39] Zhong W, Tzounas G, Liu M, Milano F. On-line inertia estimation of virtual power plants. Electr Power Syst Res 2022;212:108336. http://dx.doi.org/10.1016/j.epsr.2022.108336.

[40] Teager HM, Teager SM. Evidence for nonlinear sound production mechanisms in the vocal tract. In: Speech production and modelling, Vol. 55. ser. NATO Advanced Study Institute Series D, W. J. Hardcastle and A. Marchal, Eds. Boston, MA: Kluwer; 1989, p. 17–29.

[41] Kaiser J. On a simple algorithm to calculate the 'energy' of a signal. In: International conference on acoustics, speech, and signal processing, vol. 1. 1990, p. 381–4. http://dx.doi.org/10.1109/ICASSP.1990.115702.

[42] Kamwa I, Pradhan AK, Joós G. Robust detection and analysis of power system oscillations using the Teager-Kaiser energy operator. IEEE Trans Power Syst 2010;26(1):323–33.

[43] Maragos P, Potamianos A. Higher order differential energy operators. IEEE Signal Process Lett 1995;2(8):152–4.

[44] Maragos P, Kaiser JF, Quatieri TF. Energy separation in signal modulations with application to speech analysis. IEEE Trans Signal Process 1993;41(10):3024–51.

[45] de la O Serna JA, Ramirez JM, Zamora Mendez A, Paternina MRA. Identification of electromechanical modes based on the digital Taylor-Fourier transform. IEEE Trans Power Syst 2016;31(1):206–15. http://dx.doi.org/10.1109/TPWRS.2015.2403290.

[46] de la O Serna JA, Paternina MA, Zamora-Mendez A. Assessing synchrophasor estimates of an event captured by a phasor measurement unit. IEEE Trans Power Deliv 2021;36(5):3109–17. http://dx.doi.org/10.1109/TPWRD.2020.3033755.

[47] Serna JAdlO. Dynamic Harmonic analysis with FIR filters designed with O-splines. IEEE Trans Circuits Syst I Regul Pap 2020;67(12):5092–100. http://dx.doi.org/10.1109/TCSI.2020.2996976.

[48] Kundur P, Balu NJ, Lauby MG. Power system stability and control, Vol. 7. McGraw-hill New York; 1994.

[49] Zhao J, Tang Y, Terzija V. Robust online estimation of power system center of inertia frequency. IEEE Trans Power Syst 2019;34(1):821–5. http://dx.doi.org/10.1109/TPWRS.2018.2879782.

[50] Sauer PW, et al. Power system dynamics and stability with synchrophasor measurement and power system toolbox. Second Ed.. John Wiley & Sons; 2017.

[51] Canizares C, et al. Benchmark models for the analysis and control of small-signal oscillatory dynamics in power systems. IEEE Trans Power Syst 2017;32(1).

[52] Enhancing wide-area damping controllers via data-assisted power system linear models. Electr Power Syst Res 2023;217:109085. http://dx.doi.org/10.1016/j.epsr.2022.109085.